KAIZEN MATHS

Lesson starter

Percentage Fluency

Advanced Percentage Applications

Name _____

S1

104 is increased by 14%, then decreased by 20%. Find the final amount.

S2

132 is 93% of the original amount. Find the original amount.

S3

An item costs £143. A 10% discount is applied, then 15% VAT is added. Find the final price.

S4

174 is increased by 13%, then decreased by 17%. Find the final amount.

KAIZEN MATHS

Lesson starter

Percentage Fluency

Advanced Percentage Applications

Name _____

S1

104 is increased by 14%, then decreased by 20%. Find the final amount.

S2

132 is 93% of the original amount. Find the original amount.

S3

An item costs £143. A 10% discount is applied, then 15% VAT is added. Find the final price.

S4

174 is increased by 13%, then decreased by 17%. Find the final amount.

NUM-4608-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Percentage Fluency

Advanced Percentage Applications

Name _____

E1

97 is 63% of the original amount. Find the original amount.

E2

An item costs £121. A 30% discount is applied, then 18% VAT is added. Find the final price.

E3

257 is increased by 23%, then decreased by 8%. Find the final amount.

KAIZEN MATHS

Exit ticket

Percentage Fluency

Advanced Percentage Applications

Name _____

E1

97 is 63% of the original amount. Find the original amount.

E2

An item costs £121. A 30% discount is applied, then 18% VAT is added. Find the final price.

E3

257 is increased by 23%, then decreased by 8%. Find the final amount.

NUM-4608-L4 · E

Show your working

KAIZEN MATHS

Exit ticket

Percentage Fluency

Advanced Percentage Applications

Name _____

E1

97 is 63% of the original amount. Find the original amount.

E2

An item costs £121. A 30% discount is applied, then 18% VAT is added. Find the final price.

E3

257 is increased by 23%, then decreased by 8%. Find the final amount.

NUM-4608-L4 · E

Show your working

KAIZEN MATHS

Exit ticket

Percentage Fluency

Advanced Percentage Applications

Name _____

E1

97 is 63% of the original amount. Find the original amount.

E2

An item costs £121. A 30% discount is applied, then 18% VAT is added. Find the final price.

E3

257 is increased by 23%, then decreased by 8%. Find the final amount.

NUM-4608-L4 · E

Show your working

Teacher answers & working

Percentage Fluency · Advanced Percentage Applications

NUM-4608-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

104 is increased by 14%, then decreased by 20%. Find the final amount.

Answer:

94.85

Working:

First multiplier = 1 + 14 ÷ 100 = 1.14

Apply the 14% increase: 104 × 1.14 = 118.56

Second multiplier = 1 − 20 ÷ 100 = 0.8

Apply the 20% decrease to the new amount: 118.56 × 0.8 = 94.85

Combined calculation: 104 × 1.14 × 0.8 = 94.85

Therefore, final amount = 94.85.

S2

132 is 93% of the original amount. Find the original amount.

Answer:

141.94

Working:

Let the original amount be x.

$93\%\ \text{of}\ x = 132$

$0.93 \times x = 132$

$x = 132 \div 0.93$

$x = 141.94$

Therefore, original amount = 141.94.

S3

An item costs £143. A 10% discount is applied, then 15% VAT is added. Find the final price.

Answer:

£148.01

Working:

Discount multiplier = 1 − 10 ÷ 100 = 0.9

Price after discount = £143 × 0.9 = £128.7

VAT multiplier = 1 + 15 ÷ 100 = 1.15

Final price = £128.7 × 1.15 = £148.01

Combined calculation: £143 × 0.9 × 1.15 = £148.01

Therefore, final price = £148.01.

Teacher answers & working

Percentage Fluency · Advanced Percentage Applications

NUM-4608-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

174 is increased by 13%, then decreased by 17%. Find the final amount.

Answer:

163.19

Working:

First multiplier = 1 + 13 ÷ 100 = 1.13

Apply the 13% increase: 174 × 1.13 = 196.62

Second multiplier = 1 − 17 ÷ 100 = 0.83

Apply the 17% decrease to the new amount: 196.62 × 0.83 = 163.19

Combined calculation: 174 × 1.13 × 0.83 = 163.19

Therefore, final amount = 163.19.

E1

97 is 63% of the original amount. Find the original amount.

Answer:

153.97

Working:

Let the original amount be x.

$63\%\ \text{of}\ x = 97$

$0.63 \times x = 97$

$x = 97 \div 0.63$

$x = 153.97$

Therefore, original amount = 153.97.

E2

An item costs £121. A 30% discount is applied, then 18% VAT is added. Find the final price.

Answer:

£99.95

Working:

Discount multiplier = 1 − 30 ÷ 100 = 0.7

Price after discount = £121 × 0.7 = £84.7

VAT multiplier = 1 + 18 ÷ 100 = 1.18

Final price = £84.7 × 1.18 = £99.95

Combined calculation: £121 × 0.7 × 1.18 = £99.95

Therefore, final price = £99.95.

Teacher answers & working

Percentage Fluency · Advanced Percentage Applications

NUM-4608-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

257 is increased by 23%, then decreased by 8%. Find the final amount.

Answer:

290.82

Working:

First multiplier = 1 + 23 ÷ 100 = 1.23

Apply the 23% increase: 257 × 1.23 = 316.11

Second multiplier = 1 − 8 ÷ 100 = 0.92

Apply the 8% decrease to the new amount: 316.11 × 0.92 = 290.82

Combined calculation: 257 × 1.23 × 0.92 = 290.82

Therefore, final amount = 290.82.